

BEHIND THE SETS

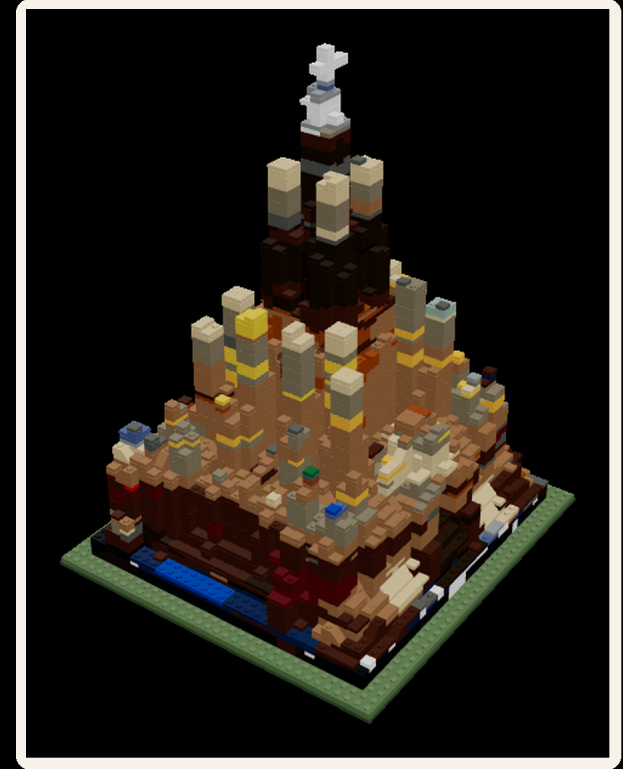


LEgoarCh

Name a building → get a buildable LEGO set.

The engineering under the demo → how a sentence becomes real, orderable bricks.

Generative AI · Emilie El Chidiac & Charles Abi Chahine · 2026





Computation you can hold.

Three reasons a generated set beats a render.



Real, not rendered

Generative form that snaps together in real bricks.

FOR THE COMPUTATIONAL DESIGNER



Models, not mockups

Hand a client a set they keep → not foam they bin.

FOR THE PRACTICING ARCHITECT



Iterate in bricks

Change the design → reorder only what changed.

FOR THE STUDENT & MAKER

A form a machine dreamed up → that real bricks can actually build.



One pipeline, end to end.

Five moves turn a sentence into an orderable set.



GENERATIVE AI · proposes the form

DETERMINISTIC CODE · proves & orders

Anyone can generate a picture → the back half, the code that proves it builds, is the work.



What you actually do.

One sentence in → a real set on your shelf. Five moves, every one yours to redo.



Every stop is editable → re-roll the render, re-tune the bricks, reopen any set from the shelf.



One dreams → one carves.

Two AI models do all the imagining. Everything after is plain code.



FLUX.2 Klein 4B

base image model → dreams the picture



+ legoarch LoRA · 40 real Architecture sets



TRELLIS.2-4B

render → carves a textured 3D mesh



invents the faces the photo can't see

qwen3-4b encoder · flux2-vae · euler · 1024×1024

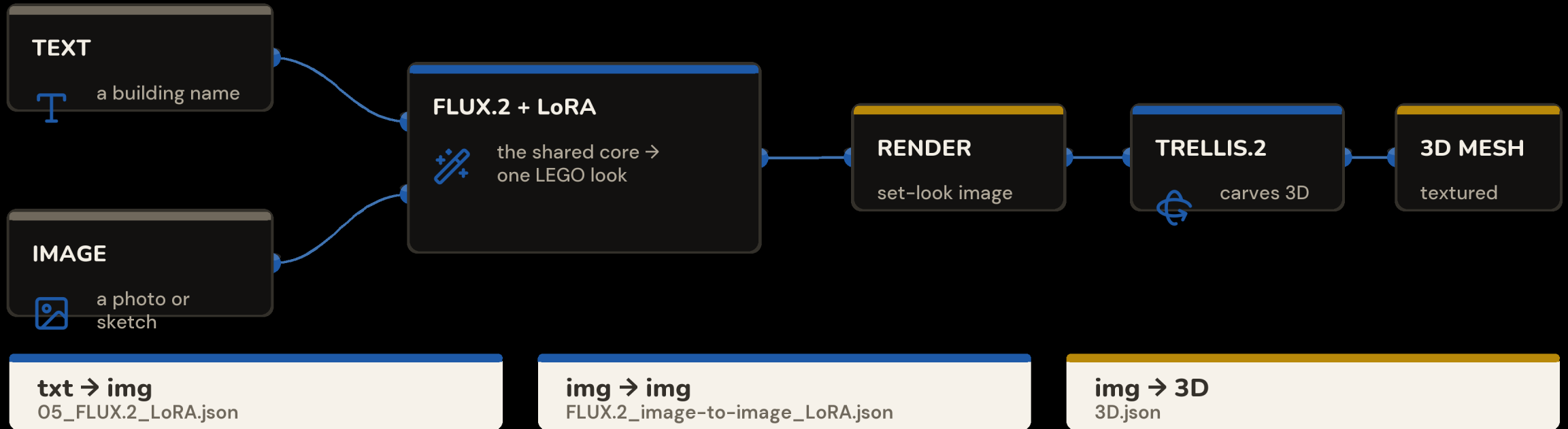
swept, then fixed → 28 steps · CFG 5.0 · LoRA 1.0 · negative prompt on

Two models dream → everything after them is plain, checkable code.



Three graphs, one core.

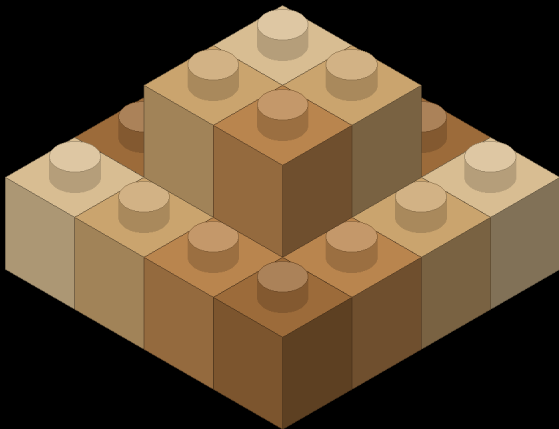
Text or image in → the same FLUX.2 + legoarch LoRA core → then TRELIS to 3D.



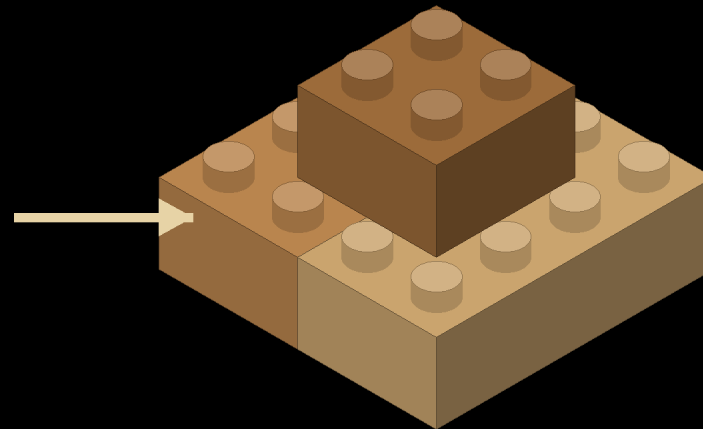


Many 1×1s → a few real parts.

Split-and-merge packs the voxel grid into the largest legal bricks — fewer parts, same shape.



raw voxel grid
thousands of 1×1 cells



merged real parts
 $2 \times 4 \cdot 2 \times 2 \cdot \text{slopes}$

FOUR PASSES

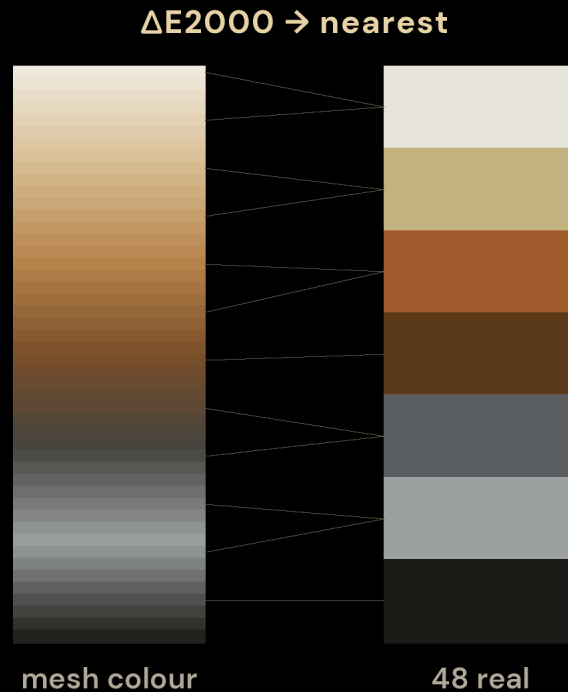
-  slopes
-  bricks
-  plates
-  tiles

Every brick is colour-uniform and seam-staggered → every footprint is one a real mould was made in.



Every voxel → a real colour.

A continuous mesh colour is snapped to the nearest of 48 real LEGO colours by ΔE_{2000} .



- 1 Sample**
read each voxel's colour off the textured mesh
- 2 Match**
rescale exposure to the FLUX render (per-channel quantile)
- 3 Convert**
sRGB → CIE Lab (D65)
- 4 Snap**
nearest of 48 real colours by CIEDE2000
- 5 Clamp**
to a colour that part was really moulded in

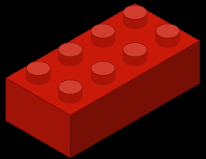
Continuous mesh colour → a small set of real, orderable LEGO colours. No invented paint.



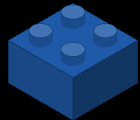
BUILDABLE MEANS ORDERABLE

Real parts. Real moulds.

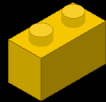
A small vocabulary of actual BrickLink parts → snapped to real colours.



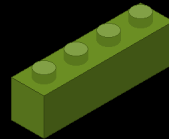
2×4
3001



2×2
3003



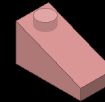
1×2
3004



1×4
3010



1×1
3005



slope
3040



48 real LEGO colours

44

real parts

1,598

validated combos



WE TESTED THIS

The look lives in the LoRA.

Same prompt, same seed → only the LoRA strength changes.

Sagrada

LoRA 0



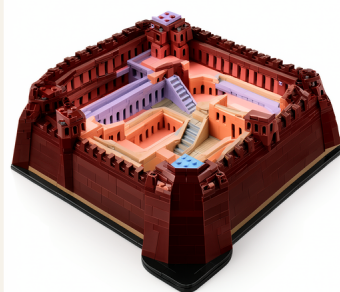
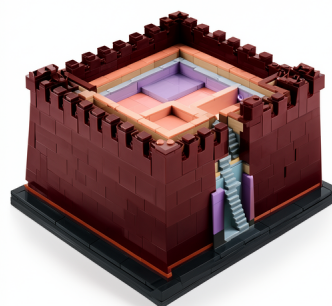
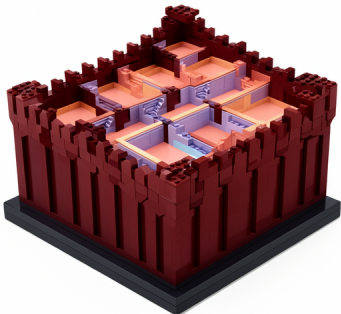
LoRA 0.5



LoRA 1.0



La Muralla



winner

0 → 0.5 → 1.0

At 0 it's a plain building. At 1 the studs and seams snap in → on a landmark and an obscure block alike.

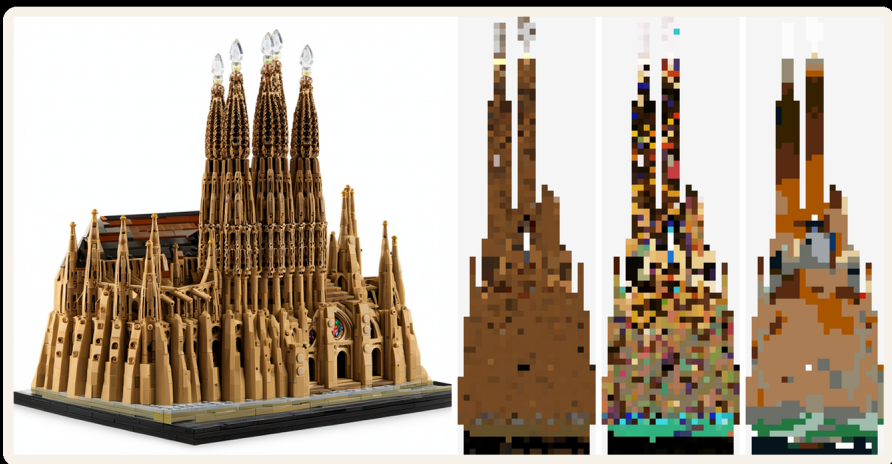
The LEGO look isn't the prompt. It's the **fine-tune.**
swept 0 → 1.5 · winner 1.0



WE TESTED THIS

Same building, -46% bricks.

Raw mesh colour speckles into thousands of 1x1s → we merge it to the true colour first.



real build → raw colour grid → merged to true colour

1x1 speckle



merged regions

SAGRADA FAMÍLIA

-46%

8,627 → 4,682 pieces

WHY FEWER IS BETTER

cheaper · simpler to build · same stability

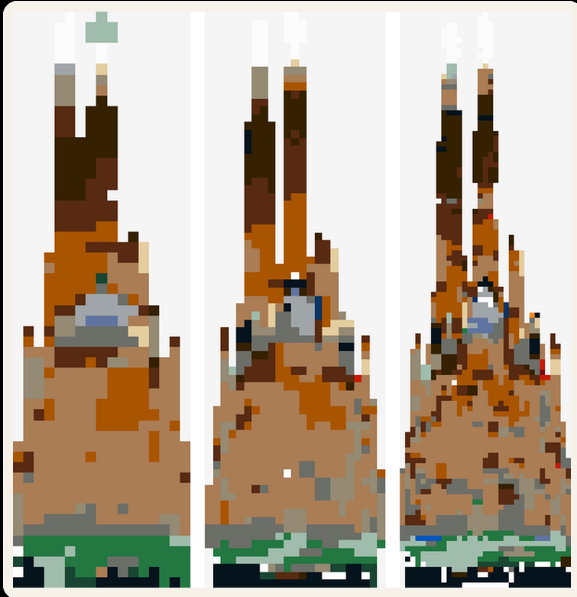
holds across buildings → -19% to -46%



WE TESTED THIS

Draft to flagship — same set.

Pick the detail; the method holds. Pieces grow about quadratically.



Sagrada Família at detail 24 · 32 · 48

detail 24

draft

2,189
pieces

detail 32

default · winner

4,682
pieces

detail 48

large

15,089
pieces

16–64 studs available in-app · same connected set every time

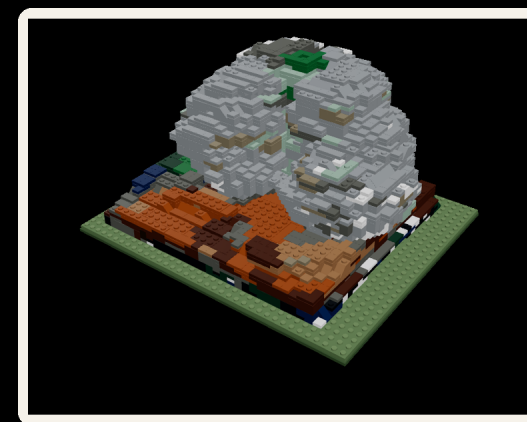


It built Gehry, not the Guggenheim.

We named both. The model followed the curves — and voxels can't hold them.

THE PROMPT — VERBATIM

"Guggenheim Museum Bilbao Frank Gehry ...
interconnected swirling titanium-clad volumes ...
overlapping curved ship-like masses merging into one
continuous sculptural body ..."

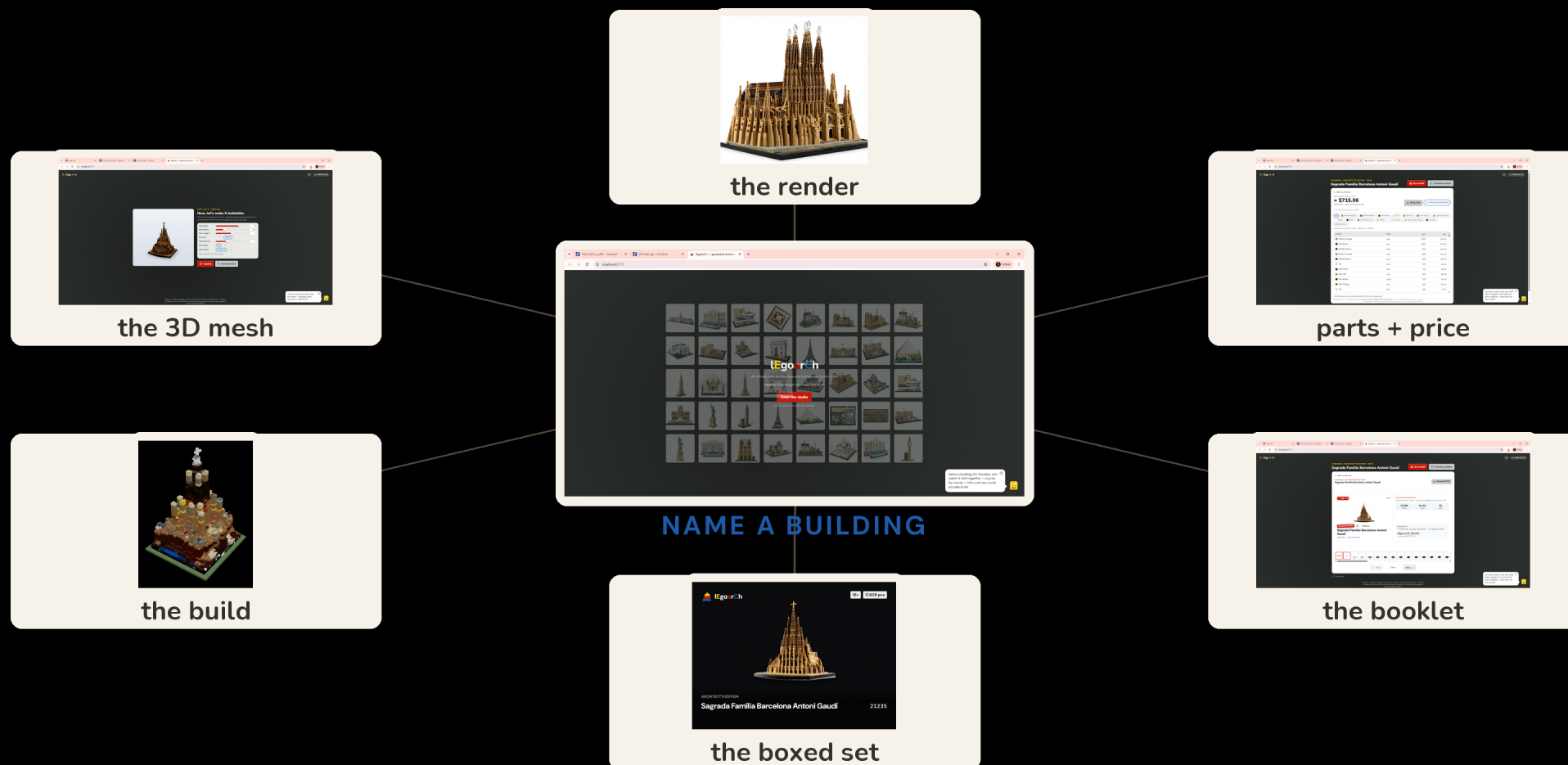


voxelized Bilbao → a connected metallic blob

Smooth curves can't survive a voxel grid.

Connected and 93% supported → just not legible. Knowing exactly where the method ends is part of the work.

One sentence → everything you get.



lEgoarCh

From words to LEGO.

Generative AI proposes → code proves → the catalog ships.

Emilie El Chidiac & Charles Abi Chahine · 2026